

Cournot's principle and Lewis's best-system interpretation of probability

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- 1 Should Cournot's principle and Lewis's best-system criterion be thought of as competing definitions of objective probability?
- 2 What are the advantages and disadvantages of Cournot's principle, compared with Lewis's high probability, as a measure of the fit of a probabilistic theory to the world?
- 3 When, if ever, is Lewis's notion of "perfectly natural" properties and relations needed for Cournot's principle?
- 4 Cournot's principle is vague; we must decide how high a probability is needed for practical certainty and which events with high probability are sufficiently simple or remarkable. Borel argued that this indeterminacy reflects the nature and diversity of probability's applications. Do you agree?

Glenn's questions vs. my plan

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Role Cournot & Lewis play in statistical inference?

- Data Y with unknown distribution P
- Quantity of interest: $\theta = \theta(P)$ in Θ
- No prior information about θ
- Goal is (probabilistic) inference on θ :
 - construct a data-dependent “probability”² $\bar{\Pi}_Y$ on Θ
 - if $\bar{\Pi}_Y(A)$ is small, then infer A^c
- Must be “reliable” in some sense
- This was Fisher’s goal, the *Holy Grail* (Efron)

²Actually, a lower/upper probability pair $(\underline{\Pi}_Y, \bar{\Pi}_Y)$...

- Two “probabilities” in play:
 - chance, P — whatever interpretation
 - credence, $\bar{P}_Y$ — subjective, Y -dependent beliefs
- Need some (minimal) link between the two...
- Validity: *the chance that my credence leads to an erroneous inference should be (controllably) small*
- Validity and Cournot & Lewis?

- $\bar{\Pi}_Y$ is subjective, internal — relevant to real-world?
- At a minimum:
 - $\bar{\Pi}_Y$ shouldn't be externally meaningless
 - i.e., " $\bar{\Pi}_Y(A)$ small" should make others doubt A
- Want something like Cournot's principle:

$$\bar{\Pi}_Y(A) \text{ small} \implies \text{others agree probably } A \not\approx \theta$$

- Can't directly justify this, need an indirect path through properties of my belief *construction*, $Y \mapsto \bar{\Pi}_Y \dots$

- Suppose we have

“ $\bar{\Pi}_Y(A)$ small” is a P-rare event for all $A \ni \theta$

- Fisher:

$\bar{\Pi}_Y(A)$ small $\implies A \not\ni \theta$ or a P-rare event happened

- Cournot: P-rare events rarely happen
- Roundabout path to what we wanted, i.e.,

$\bar{\Pi}_Y(A)$ small $\implies$ others agree probably $A \not\ni \theta$

- Above analysis focused on linking chance and credence for certain cases — “small $\overline{\Pi}_Y(A)$ ” and “P-rare events”
- Lewis’s *Principal Principle* doesn’t distinguish between small and not-small chance/credence
- A version of *PP* in this context: more extensive calibration of $\overline{\Pi}_Y$ and P covering small & not-small magnitudes?

- A simple/artificial interpretation:
 - $\overline{\Pi}_Y$ gives (bounds on) prices one's willing to pay for a bet
 - internal & external coherence
- Lewis's *best-system* interpretation considers a balance of simplicity, strength, and fit
- What's "best" for $\overline{\Pi}_Y$ -type systems?
- Efficiency:
 - "small" $\overline{\Pi}_Y$ means more efficiency, e.g., more power
 - make $\overline{\Pi}_Y$ as small as possible without ruining validity
- Interpretation: my $\overline{\Pi}_Y$ -beliefs are what they are because my $\overline{\Pi}_Y$ is valid & efficient

Validity condition — link between chance & credence

$$P\{\overline{\Pi}_Y(A) \leq \alpha\} \leq \alpha, \quad \text{all } \alpha \in [0, 1] \text{ and all } A \ni \theta(P).$$

- I've been working on credence-assignment rules $Y \mapsto \overline{\Pi}_Y$ that satisfy the above condition
- There are foundational questions to consider
- Ideas of Cournot & Lewis (and others) are clearly relevant
- It'd be fun to dig into this more